Finding Images Through Spoken Language

A simple project overview — Flickr8k Audio + image retrieval

Core idea
Say what you are looking for.
The system finds matching pictures.

What are we building?

A search system that understands a spoken description without first turning it into text.

1

You speak

“A child playing with a dog”

2

AI compares meaning

Speech becomes a vector; each image also has a vector.

3

Pictures returned

Show the top matching images, ranked by similarity.

Important: The main approach searches directly from audio. Speech recognition can be a separate baseline, not the core retrieval path.

Why Flickr8k Audio?

It gives us spoken descriptions paired with the images they describe.

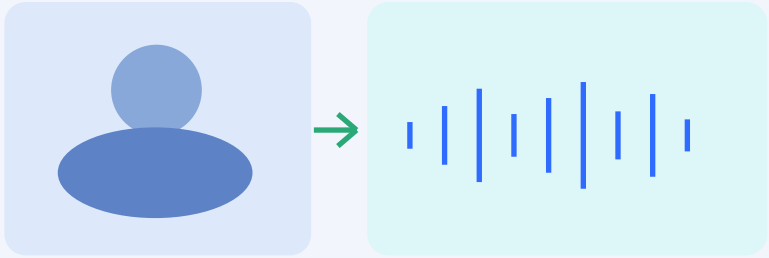


IMAGE SPOKEN CAPTION

A training example = one image + one spoken caption that describes it.

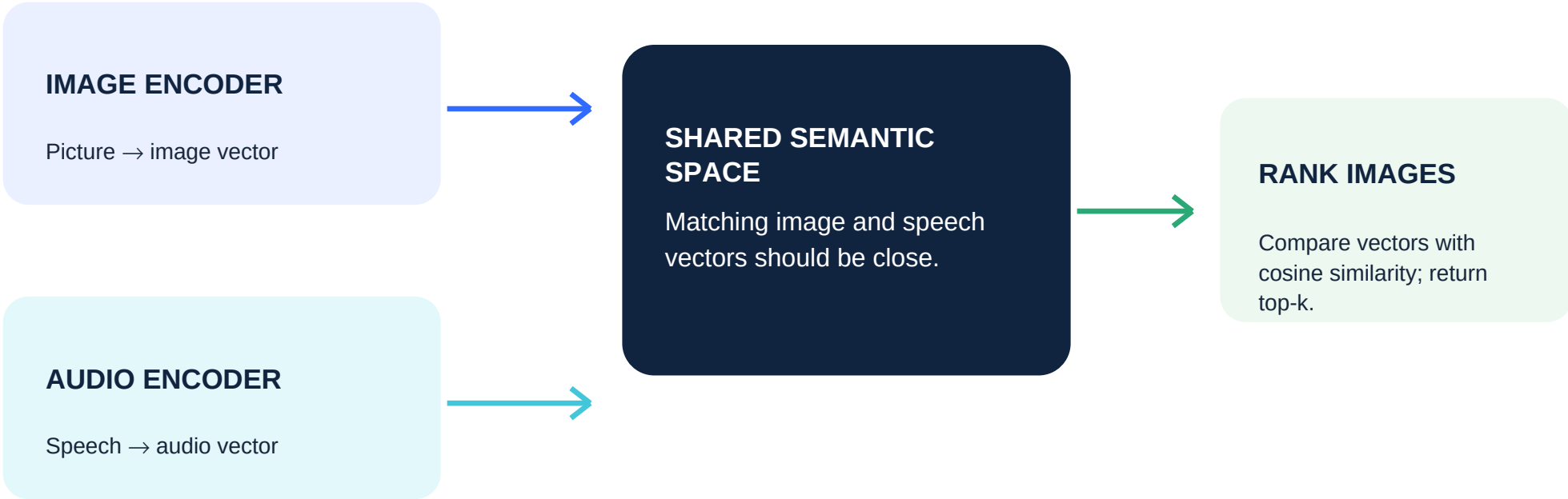
What we must verify

- Download the spoken-caption files and matching Flickr8k images.
- Check which audio files map to which image IDs.
- Measure the real usable audio duration and inspect missing/corrupt files.
- Keep all captions for the same image in one data split.

Dataset page: sls.csail.mit.edu/downloads/flickraudio/index.cgi

Two encoders, one shared space

Each encoder turns a different kind of input into a vector that can be compared.



Training: matched pairs are positives; other items in the batch provide negatives. A contrastive loss teaches the encoders to align.

How training works

Start with pretrained encoders and small projection layers; keep the first version simple.

1

Build pairs

Load an image and its matching audio caption.

2

Encode

Create one vector for the image and one for the audio.

3

Compare

Give matching pairs high similarity; mismatches lower similarity.

4

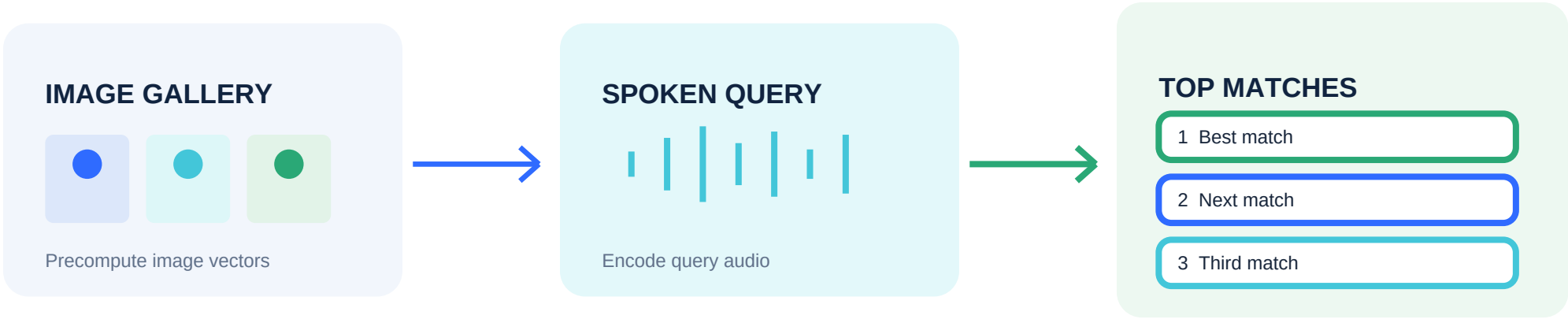
Update

Use contrastive loss to improve the shared space.

Data split rule: Split by image ID before training. Captions for the same image must not leak into both train and test sets.

How a search happens

The expensive image encoding is done once and cached.



At query time: audio vector → similarity scores → ranked images. No need to re-encode the whole gallery for every search.

06 • EVALUATION

How will we know it works?

Measure retrieval quality on images that were not used for training.

Recall@1

Was the correct image ranked first?

Recall@5 / @10

Was it in the top five or ten?

Similarity check

Do true pairs score higher than mismatched pairs?

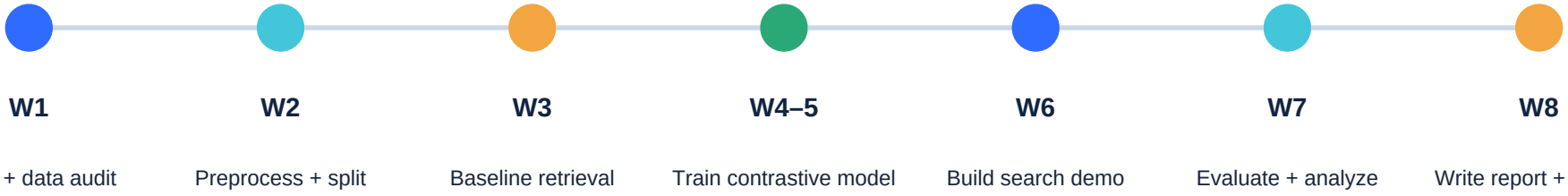
Generalization

Does it work for new images or alternative captions?

One-shot / few-shot: Agree on the exact setup with the supervisor before running experiments. Report results on held-out images.

A simple 8-week roadmap

Aim for a working baseline early, then improve and evaluate.



Minimum successful result: a reproducible model, a working audio-to-image search demo, and clear test metrics — even if retrieval quality is modest.

What to do first

Reduce uncertainty before spending time on model training.

1	**Get the data**	Download audio and corresponding Flickr8k images; check access and licence.
2	**Verify the pairs**	Confirm every audio caption links to the correct image ID.
3	**Measure the data**	Count usable files, durations, sample rates and missing items.
4	**Confirm the protocol**	Ask the supervisor about encoders, compute, and one-/few-shot evaluation.

Data links

Flickr8k Audio: https://sls.csail.mit.edu/downloads/flickraudio/index.cgi

Project note: raw datasets and large checkpoints should stay out of Git.